Report: Kraken2 vs KrakenUniq - Family-level Taxonomic Comparison

This report presents the results of a comparison between two metagenomic classification pipelines, Kraken2 (GTDB database) and KrakenUniq (MicrobialNT database), applied to ancient DNA samples from archaeological necropolises in the Valencia area, spanning Roman, Visigothic, Islamic, Medieval, and Christian periods.

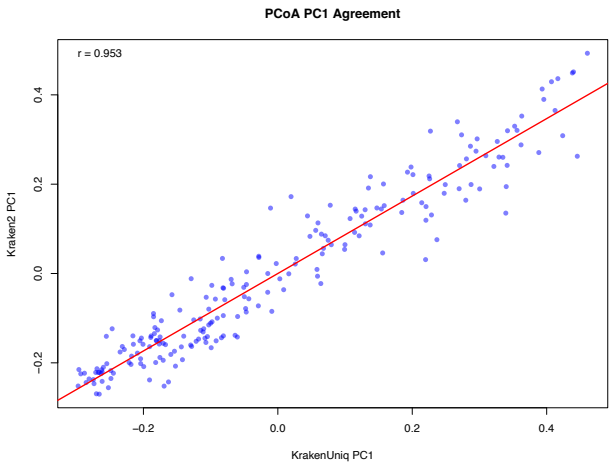
Dataset. Family-level abundance matrices were built from the classification reports of both tools. Samples were filtered by retaining only those present in the project metadata file (metadataGOG.xlsx), removing controls and unrelated samples. To ensure direct comparability between the two tools, all analyses were performed on the intersection of samples classified by both: **n = 211 samples** for which both Kraken2 and KrakenUniq produced a valid microbial profile. The family Hominidae was removed from both matrices prior to analysis.

Report contents. The figures show:

- overall agreement between the two tools (PC1 Agreement, Procrustes analysis);
- concordance at the level of individual families (Spearman correlations, composition barplot, Venn diagram);
- ecological structure of the dataset (PCoA coloured by archaeological site, historical period, and bone material type).

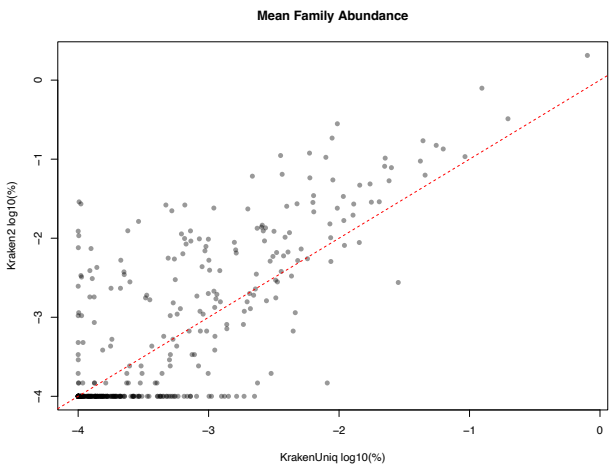
PC1 Agreement (r = 0.953)

Each point represents one sample, plotted by its first principal coordinate (PC1) value from Kraken2 (y-axis) against KrakenUniq (x-axis). The high Spearman correlation (r = 0.953) demonstrates that both tools identify the same dominant axis of variation in microbial community composition: samples that appear compositionally distinct under Kraken2 are almost invariably distinct under KrakenUniq as well. This result is consistent with the Procrustes statistic ($r \approx 0.90$, $p < 0.001$) and confirms that at the level of overall beta-diversity, the two pipelines reach equivalent biological conclusions despite using different algorithms and reference databases.



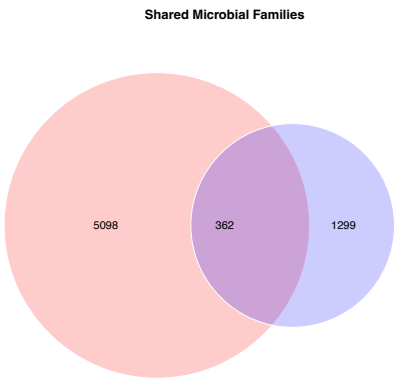
Mean Family Abundance

Each point represents one of the 362 shared taxonomic families, plotted as its mean relative abundance in Kraken2 (y-axis) vs KrakenUniq (x-axis), both on a log10 scale. The dashed red line is the 1:1 diagonal (= perfect agreement). Most points fall above the diagonal, indicating that Kraken2 systematically assigns higher relative abundances to the same families compared to KrakenUniq. The dense cluster near the detection floor maybe represents families detected by one tool but virtually absent in the other. The wide scatter around the diagonal confirms that even among shared families, abundance estimates are not directly interchangeable between the two tools, a key caveat for any taxon-specific downstream analysis.



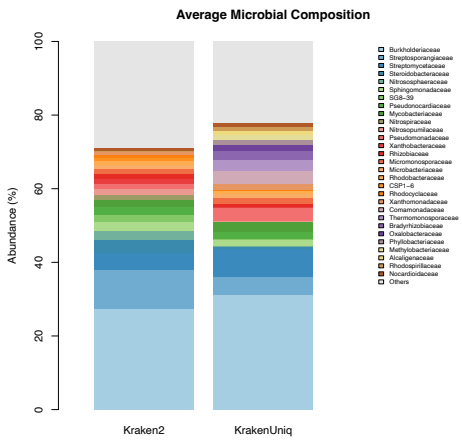
Venn Diagram: Shared Microbial Families

The diagram illustrates the taxonomic overlap between the two tools at family level. Of all families detected, only 362 are shared, while Kraken2 identifies 5,098 exclusive families and KrakenUniq 1,299. This low overlap is not primarily a biological disagreement: it reflects the nomenclature gap between the two reference databases. Kraken2's GTDB uses alphanumeric placeholders for many bacterial lineages not yet formally named, while KrakenUniq's MicrobialNT follows standard NCBI taxonomy and additionally includes eukaryotes, fungi, and viruses absent from GTDB. The same organism may therefore appear under a different or unrecognised name in each database, making direct family-level comparison unreliable without prior name harmonisation.



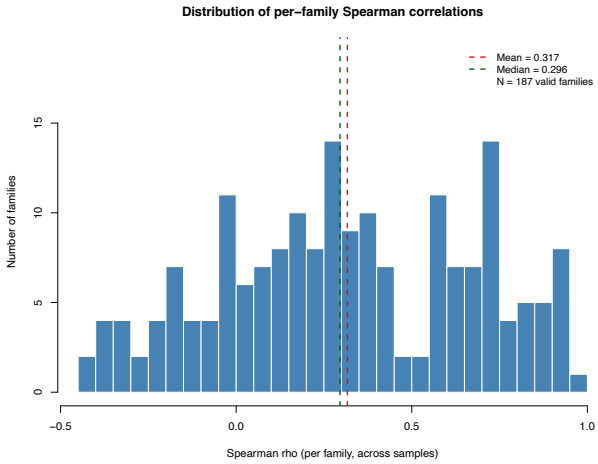
Average Microbial Composition

The stacked barplot shows the average relative abundance of the top 20 families, normalised to 100%, for Kraken2 and KrakenUniq side by side. Despite the low taxonomic overlap shown in the Venn diagram, the dominant families are consistent: Burkholderiaceae is the most abundant taxon in both tools, followed by members of Actinobacteria and Proteobacteria typical of soil environments. This soil-associated profile is expected for ancient bone samples that have absorbed the burial sediment microbiota over centuries. Proportional differences between the two bars reflect database-driven variation in classification rather than true biological differences in community composition.



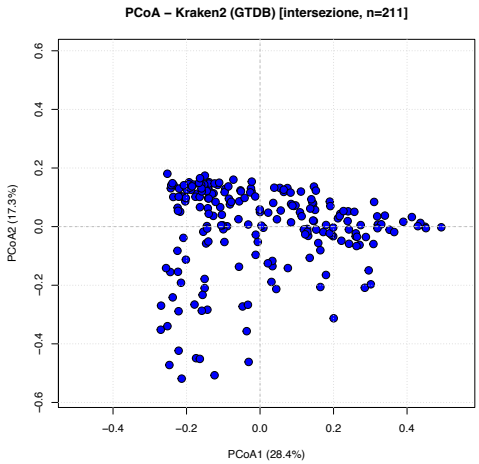
Distribution of per-family Spearman Correlations

The histogram summarises the per-family Spearman correlation (ρ) between Kraken2 and KrakenUniq across all shared families. The distribution is broad, ranging from strongly negative to near-perfect agreement, with a mean ρ ≈ 0.31 and median ρ ≈ 0.30. Only few of families show strong concordance (ρ > 0.8), while a substantial fraction shows weak or negative correlations. The few highly consistent families are the most abundant ones; abundant taxa are sampled deeply enough to be reliably detected by both tools regardless of database differences. This graph is the clearest statistical expression of the "dual reality": strong global agreement coexisting with low concordance at the level of individual taxa, justifying caution when interpreting taxon-specific results.



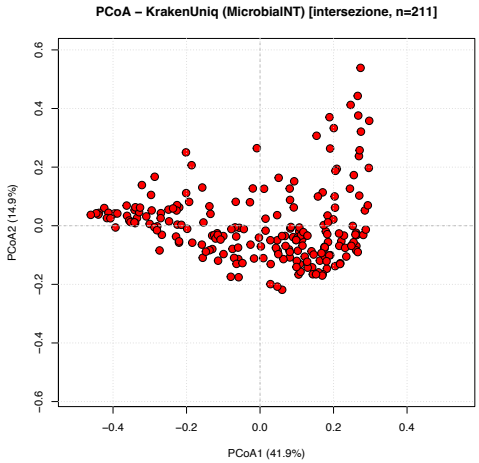
PCoA Kraken2 (uniform)

Overall distribution of the 211 samples in PCoA space based on Bray-Curtis dissimilarity calculated on Kraken2 relative abundances. PC1 explains 28.4% of total variance and PC2 17.3%. The point cloud does not show discrete clusters, suggesting a continuous gradient of microbial community variation across samples rather than clearly separated groups. This plot serves as a baseline before introducing metadata variables.



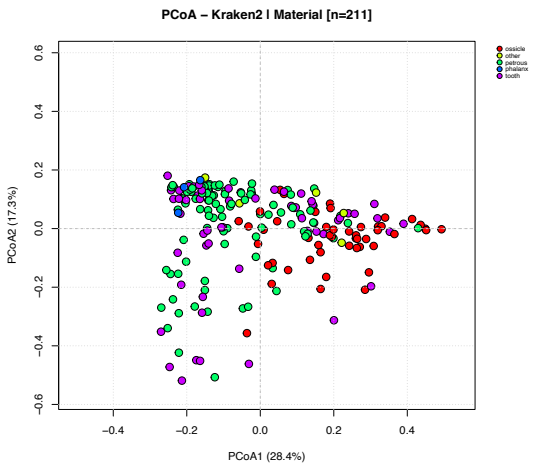
PCoA KrakenUniq (uniform)

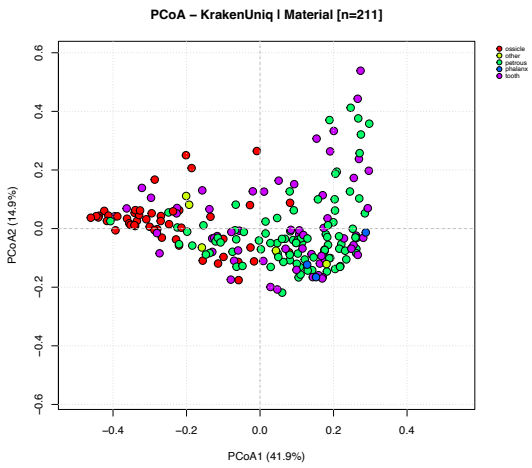
Same ordination as above, computed from KrakenUniq relative abundances. PC1 explains 41.9% of total variance and PC2 14.9%, notably higher than Kraken2, suggesting that KrakenUniq's broader database (MicrobialNT, including fungi, viruses and eukaryotes) captures a stronger primary gradient in the data. The overall shape of the point cloud is broadly consistent with Kraken2, visually confirming the high global agreement (r = 0.953) quantified in the PC1 Agreement plot.



PCoA Kraken2 | Material and PCoA KrakenUniq | Material

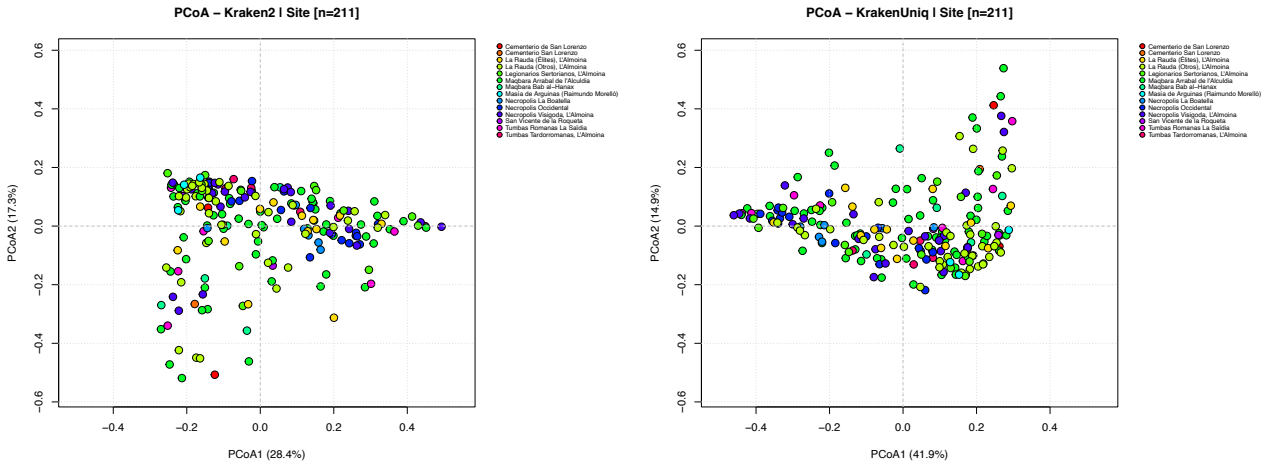
Points are coloured by bone material type (ossicle, petrous, phalanx, tooth, other). A visible tendency emerges in both tools: ossicles (the small, highly porous bones of the middle ear) concentrate at positive PC1 values, partially separating from the remaining material types. This likely reflects a taphonomic effect: the physical structure of the ossicle favours different rates of environmental microbial infiltration compared to denser cortical bones. The consistency of this pattern across both tools strengthens the conclusion that it represents a genuine biological signal rather than a classification artefact. Material type therefore appears to be a relevant predictor of microbial community composition and should be accounted for in downstream analyses.

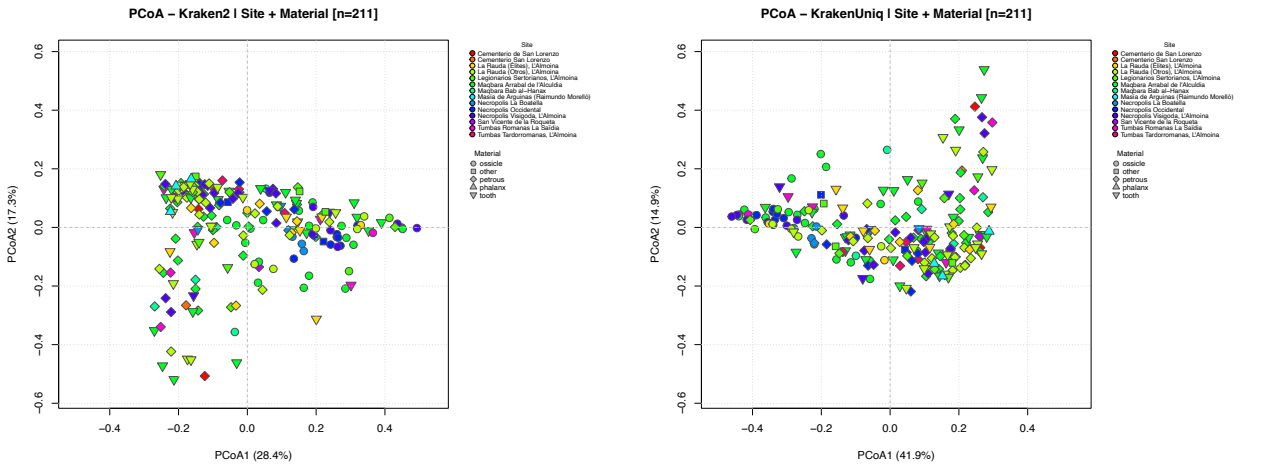




PCoA Kraken2 | Site and PCoA KrakenUniq | Site

Points are coloured by archaeological site (~14 necropolis sites). No clear geographic clustering is visible: samples from the same site are scattered throughout the ordination space in both tools. This is a biologically coherent result in archaeometagenomic studies: the post-mortem microbial signal in ancient bones is largely determined by burial soil conditions and individual preservation state, rather than by the geographic proximity of sites within the same metropolitan area. The absence of site-level clustering suggests that local environmental differences between necropolises are not the primary driver of microbial community composition in this dataset.



PCoA Kraken2 | Site + Material and PCoA KrakenUniq | Site + Material

These plots encode site as point colour and material type as point shape, allowing both variables to be read simultaneously. They confirm and integrate the patterns observed in the individual plots: no geographic clustering by site (colours remain mixed throughout), but a consistent tendency for certain material types (particularly ossicles) to occupy distinct regions of the ordination space. The convergence of this pattern across both Kraken2 and KrakenUniq reinforces the conclusion that bone material type is the primary structural variable in this dataset, and that the observed variation is biological rather than an artefact of the classification pipeline.